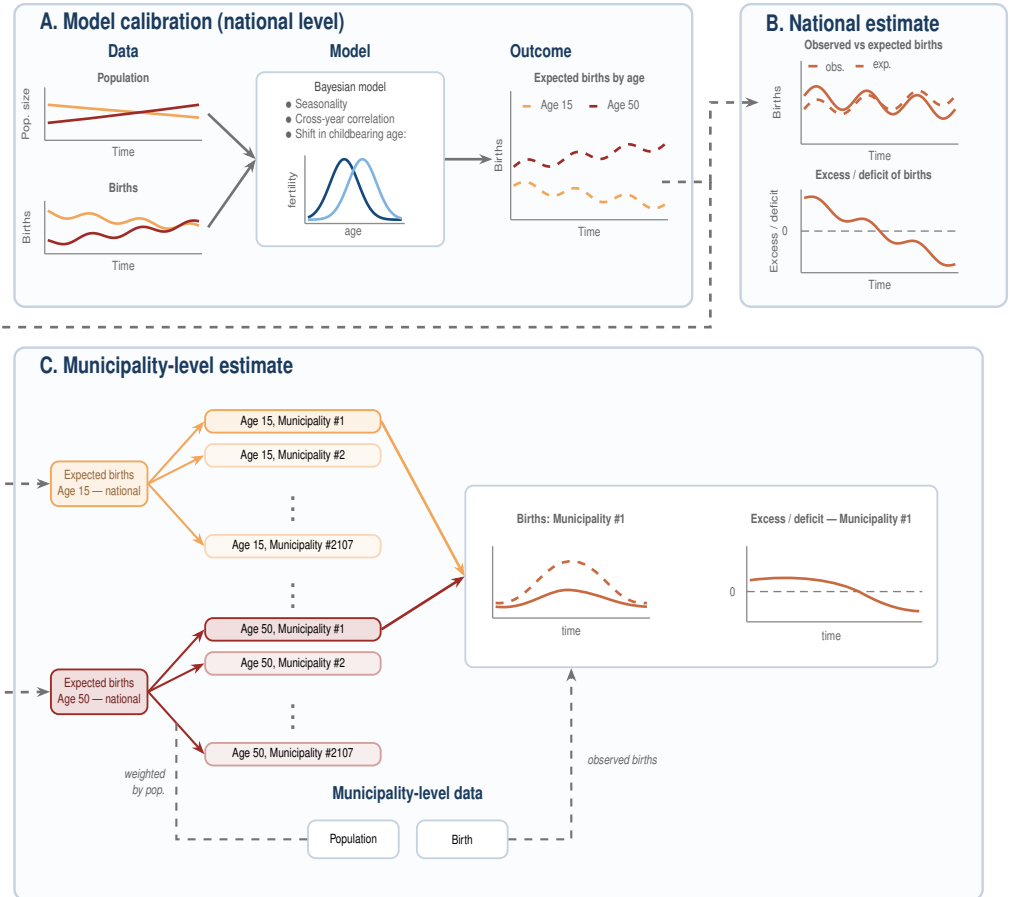

A. Model calibration (national level)
Data
Population
Pop. size
Time
Births
Births
Time
Model
Bayesian model
Seasonality
Cross-year correlation
Shift in childbearing age:
fertility
age
Outcome
Expected births by age
Age 15
Age 50
Births
Time
B. National estimate
Observed vs expected births
obs.
exp.
Births
Time
Excess / deficit of births
Excess / deficit
0
Time
C. Municipality-level estimate
Expected births Age 15 — national
Age 15, Municipality #1
Age 15, Municipality #2
Age 15, Municipality #2107
Expected births Age 50 — national
Age 50, Municipality #1
Age 50, Municipality #2
Age 50, Municipality #2107
weighted by pop.
Births: Municipality #1
time
Excess / deficit — Municipality #1
0
time
observed births
Municipality-level data
Population
Birth